A Parity Obstruction to the Exact Tseitin Width Claim Conjecture 00000003992

idealistichacker
AI-assisted implementation; no affiliation claimed

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Abstract

On the connected triangle with original vertex charges all equal to one, the standard Tseitin parity CNF has resolution width exactly two. Every subgraph of that actual charged graph with odd inherited charge has an odd number of vertices, and hence never two vertices. This refutes even the existence required by equality of width with the smallest odd-charge-subgraph size. The obstruction is independent of additional connectedness, inducedness or empty-boundary restrictions on admissible subgraphs. The pinned Lean project connects the actual graph, incidence parity and generated CNF, proves soundness of the resolution calculus, gives a width-two refutation and excludes all lower-width derivations, then proves the all-subgraph parity obstruction and necessary source clause's negation.

1 Source clause and conventions

Both exact source languages define a Tseitin contradictory system on a connected charged graph and claim its resolution width equals the vertex count of its smallest odd-charge subgraph, without an additive correction. We use the ordinary parity CNF encoding and resolution width: the maximum number of distinct literals in any used input or derived clause. Unused inputs need not occur. Resolution and weakening both require previously derived parents. An explicit finite derivation unfolds into our inductive proof trees; conversely those trees can be topologically listed as proofs. Sharing or unrolling parents does not increase width.

Charge on a subgraph means the sum modulo two of its inherited original vertex labels. The source does not specify an admissible subgraph subclass. We do *not* select one: if exact equality to a smallest member holds for any nonempty such subclass, an odd-charge subgraph of vertex count equal to the width must exist. We refute this weaker necessary existence clause for *all actual subgraphs*. An empty admissible class has no smallest member and cannot supply the asserted equality. Arbitrarily recharging a boundary-conditioned formula or replacing CNF resolution by direct XOR inference changes the input or proof system and is not silently introduced here.

2 Actual graph, parity equations and generated CNF

Let $G = K_3$, with vertices $0, 1, 2$, indexed edges $e_i = \{i, i+1 \pmod 3\}$ and charge $\chi(v) = 1$ at every vertex. This is finite, simple and connected, with total charge $1 + 1 + 1 = 1 \pmod 2$. The code uses the actual Mathlib complete graph, proves these indexed edges bijectively enumerate its unordered edge set, and identifies their incidence at each vertex with the graph's actual incidence set.

Let x, y, z be the edge variables. Its three incidence-parity equations are

$$x \oplus y = 1, \qquad y \oplus z = 1, \qquad z \oplus x = 1.$$

Their sum modulo two is $0 = 1$, so the system is contradictory. Standard parity CNF for each vertex excludes each forbidden assignment to that vertex's incident variables. A blocking clause contains the literal opposite to the assignment on every incident edge. The generated complete formula is exactly

$$F = \{x \vee y,\ \neg x \vee \neg y,\ y \vee z,\ \neg y \vee \neg z,\\ z \vee x,\ \neg z \vee \neg x\}.$$

Lean defines this generator from the actual incidence/parity data rather than merely assigning a six-clause list. It proves equality with the displayed CNF, and for *every* edge assignment proves CNF satisfaction equivalent to all vertex parity equations. The finite checks are kernel proofs of full equivalence over the actual finite domains, not an unrelated hard-coded Boolean surrogate.

3 Standard resolution width is exactly two

Clauses are finite sets of signed edge variables. Resolving C containing u with D containing $\neg u$ yields $(C \setminus \{u\}) \cup (D \setminus \{\neg u\})$. Weakening is enlargement to a superset. The bounded inductive calculus requires the cardinality of *each input, resolvent and weakened clause* to be at most the width bound. The code proves resolution soundness for every assignment, then induction proves the entire calculus sound. In particular a refutation entails actual CNF unsatisfiability.

All six inputs have width two. A derivation of width less than two cannot start: an input violates that bound, and resolution or weakening requires earlier derivations. Formally, induction on *any* purported derivation yields a contradiction at an input leaf. Thus every refutation has width at least two, not just the exhibited proof.

For the upper bound, use all inputs of width two and the following five valid steps:

Parents	Pivot	Resolvent
$x \vee y$, $\neg y \vee \neg z$	y	$x \vee \neg z$
$z \vee x$, $x \vee \neg z$	z	x
$y \vee z$, $\neg x \vee \neg y$	y	$z \vee \neg x$
$z \vee \neg x$, $\neg z \vee \neg x$	z	$\neg x$
x, $\neg x$	x	empty clause

Every line has width at most two. This gives a genuine bounded-calculus refutation. Combining it with the universal lower bound establishes the actual minimum width 2; the code also proves uniqueness of any value satisfying the minimum-width specification. We do not deduce proof width from an assigned constant or an unsatisfiability truth table.

4 All odd-charge subgraphs have odd vertex count

For *every actual Mathlib subgraph* H of K_3,

$$\sum_{v \in V(H)} \chi(v) = |V(H)| \pmod 2.$$

Since every original label is one, an odd inherited charge forces $|V(H)|$ odd. Hence no odd-charge subgraph has two vertices. The statement covers arbitrary edge subsets, connected or disconnected subgraphs, induced subgraphs and all smaller admissible classes. Ordinary subgraphs include an odd singleton of size one. If instead one requires empty boundary or a contradictory subsystem of the original equations, only the entire connected triangle can qualify; it has size three. Neither observation is needed for the stronger all-subgraph obstruction.

Theorem 1. *Conjecture 00000003992's exact width–odd-charge-subgraph-size clause is false for the connected triangle with all original charges one, under standard parity CNF resolution.*

Proof. The actual width is two. Exact equality to a smallest odd-charge-subgraph size requires an odd-charge subgraph with two vertices. None exists, since every such subgraph has odd vertex count. No choice of a nonempty admissible inherited-odd-charge subclass changes this parity contradiction. Therefore the full exact-equality assertion fails. □

5 Formal correspondence and reproducibility

The graph/edge/incidence bridge precedes the standard CNF generator and its full semantic equivalence. `DerivesWithin` has only bounded input, ordinary resolution and sound weakening constructors. `no_derivation_below_two` quantifies over arbitrary proof trees; `width_two_refutation` is the actual exhibited derivation. The minimum specification `IsResolutionWidth` is proved, not defined to be two.

`odd_charge_implies_odd_vertex_count` quantifies over every actual subgraph and actual vertex set, not a selected list of witnesses. `necessary_width_charge_clause_false` negates the existence consequence required by the source's exact minimum equality. `actual_graph_counterexample` packages actual graph connectivity, odd original charge, unsatisfiability, exact resolution width and the all-subgraph exclusion.

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